

MODEL 710-1 FIXED / REMOVABLE DISK DRIVE



The Model 710-1 Fixed/Removable Disk Drive provides additional program and/or data storage for the 700C/720C Advanced Programmable Calculator. This peripheral unit gives the user the capability of storing or retrieving alphabetic and numeric data under full format program control directly from the 700C or 720C.

The Model 710-1 Magnetic Disk Memory Unit consists of a fixed disk containing 131,072 characters, each character is equivalent to one program step, a removable disk containing an additional 131,072 characters, drive mechanism, and necessary controls. The entire assembly is contained in an attractive desk stand.

Capability of random access is featured on both the fixed and removable disks. Only one side of each disk is used for recording. The disk unit complements the 700C/720C magnetic tape cassette programming to provide an extremely powerful system.

SPECIFICATIONS

Mechanical

Disks per unit: 2 (1 fixed, 1 removable)
Record heads per unit: 2
Flying headpads per disk surface: 1
Disk outside diameter: 14 inches
Coating: Magnetic oxide

Storage

Data bytes per disk side:	131,072
Data bytes per unit (steps):	262,144
Data tracks per disk side:	64
Data tracks per unit:	128
Bytes per track:	2,048
700C/720C Register Equiv.:	32,768

NOTE:

1 byte consists of 8 bits plus parity bit.

Processing Speed

Data transfer rate:	20,833 bytes per second
Parallel data byte rate:	48 μ sec per byte avg.
Average latency:	50 msec. @ 60 Hz
	60 msec. @ 50 Hz

Position Seeking Time

Average stroke:	320 msec.
Track to track:	10 msec.
Track ϕ to home:	.96 sec.
Home to track ϕ :	7.68 sec.

CONTROLS

Complete controls for all disk operations are located on the switch panel mounted on the disk unit. Complete interlocks are provided to avoid accidental damage to the disk or accidental erasure of stored information. The following controls are provided:

Power ON/OFF — Power ON creates "Prime" moving record head to "Home" position.

Format Switch A — Up position links to upper removable disk. Down position links to fixed disk.

Format Switches

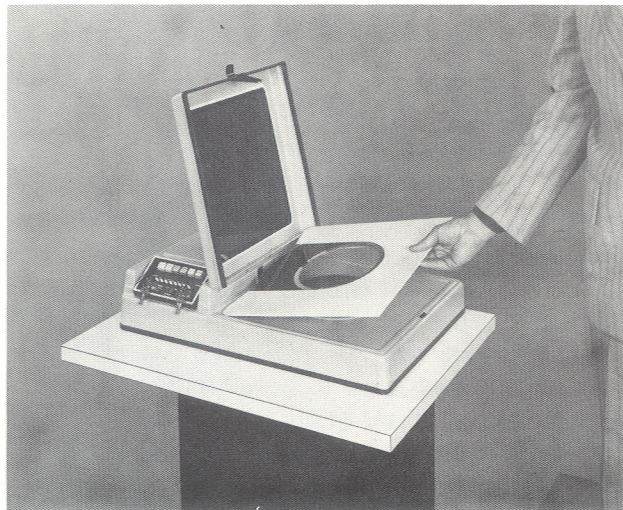
32, 16, 8, 4, 2, 1 — Provide for selection of track on either disk from 1 to 64.

Format Disk — Provides for erasure of all data and timing information on one disk surface and records new timing information.

Format Track — Provides for erasure and new timing information on a specific track.

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Copy A to B — Provides off-line transfer of all data and timing information on the removable disk to the fixed disk.

Copy B to A — Provides off-line transfer of all data and timing information on fixed disk to removable disk.

Lock Switch "A" — Prevents writing data or formatting on the removable disk.

Lock Switch "B" — Prevents writing data or formatting on the fixed disk.

PROGRAMMING

The transfer of data to or from the 700C/720C calculator is carried out by indicating the initial address of the disk in the Y-Register and the initial address of the calculator core in the X-Register and giving an operation code to the calculator. Data may be transferred in 8, 16, 32, 64, 128, 256-byte units. Single-byte units, in addition to the above, can be retrieved only.

OPERATING ENVIRONMENT

Temperature 50°F to 100°F (10°C to 48°C)
Humidity 10 - 90% relative humidity without condensation.

POWER REQUIREMENTS

115 VAC, 50/60 Hz, 3.0 amp, line cord provided to plug into wall outlet. Interconnecting cable to plug into I/O connector on 700C/720C calculator.

OVERALL DIMENSIONS & WEIGHT

Height: 32 1/2 inches
Depth: 20 3/4 inches
Width: 28 7/16 inches
Weight: 90 pounds